

Effect of environmental conditions on cetacean entanglements: a case study from the Gold Coast, Australia

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Abstract. Entanglement of marine mammals in fishing gear is recognised worldwide and is a continuous management concern. Gill-net entanglement data from the Queensland Shark Control Program (QSCP) on the Gold Coast, Australia, from 1990 to 2012 were analysed in the present study. Environmental drivers that may affect entanglements of humpback whales (*Megaptera novaeangliae*) and common dolphins (*Delphinus delphis*) were selected. *M. novaeangliae* entanglements coincided with their annual migration, with the greatest occurrences in September. *D. delphis* were mostly entangled from March to November, with the greatest occurrences in June. For both species, entanglements primarily occurred when the wave height was between 0.5 and 1.25 m, the wave power was between 0 and 5 kW m⁻¹ and the wind speed was between 12 and 19 km h⁻¹. *M. novaeangliae* entanglements were significantly more likely to occur in low rainfall (<6 mm h⁻¹), and *D. delphis* entanglements were more likely to occur during spring tides. There was a correlation between entanglements and the position of the East Australian Current's (EAC) maximum velocity, with 73% of *M. novaeangliae* entanglements and 79% of *D. delphis* entanglements occurring when the EAC's maximum velocity was west (shoreward) of its average position at 154°E. The present study provides the first set of possible management intervention targets associated with environmental conditions.

Additional keywords: *Delphinus delphis*, entanglements, environmental drivers, *Megaptera novaeangliae*, Queensland, shark nets.

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Introduction

Marine mammal entanglements

Globally, high numbers of marine mammals are entangled in fishing equipment. These entanglements are a concern for the management of long-lived species with low reproduction rates, because entanglements can lead to marked population declines (Hall *et al.* 2000). Cetacean bycatch also affects biodiversity through the removal of top predators in a marine ecosystem (Hall *et al.* 2000). In particular, gill-nets and crab pots used in fisheries are responsible for entanglements (Table 1). High numbers of entanglements have been reported from the feeding areas of humpback whales (*Megaptera novaeangliae*), such as in the Gulf of Alaska, where 71% of the observed mammals had entanglement scars (Neilson *et al.* 2009), and Newfoundland Canada, where 965 mammals had been entangled over a 30-year period (Benjamins *et al.* 2011). Various dolphin species were entrapped in fisheries, with mortalities ranging from 30 to 1000

animals in the Mediterranean and Indian Ocean depending on the type of fishery (Dawson 1991; Fertl and Leatherwood 1997; Natoli *et al.* 2008; Allen *et al.* 2014).

Entanglements of dolphins are generally fatal, with the animals drowning in the nets. Fisheries interactions resulting in dolphin bycatch can be devastating for their populations (Dawson 1991; Read *et al.* 2006). Solutions to reduce entanglements have been proposed that make the nets more acoustically visible to cetaceans, including net braiding with stainless steel or plastic tubing; however, these techniques have had limited successes (Dawson 1991). In 1994, successful trials were reported from a fishery in New Hampshire that uses acoustic alarms that emit pulses of sound and are placed on the nets to reduce the porpoise bycatch (Dawson *et al.* 1998). Subsequently, acoustic alarms were periodically trialled on the Gold Coast (Australia) between 1994 and 2000, and with improved technology between 2007 and 2013 (McPherson *et al.* 2004; Erbe and

Table 1. Overview of studies reporting cetacean entanglements in fishing equipment

Location	Year	Species	Number of entanglements	Fishing equipment	Reference
Newfoundland and Labrador, Canada	1979–2008	<i>Megaptera novaeangliae</i>	965	Gill-nets, pots, traps and rope	Benjamins <i>et al.</i> (2011)
Gulf of Maine, USA	1975–1990	<i>Megaptera novaeangliae</i>	47	Gill-nets, lobster traps, longline, weir, seine and trawls	Volgenau <i>et al.</i> (1995)
Ecuador	1996–2009	<i>Megaptera novaeangliae</i>	46	Gill-nets	Alava <i>et al.</i> (2012)
Gulf of Alaska	2003–04	<i>Megaptera novaeangliae</i>	133 (71%) of observed whales had entanglement scars	Pot gear, line and gill-nets	Neilson <i>et al.</i> (2009)
Western Australia	1990–2009	<i>Megaptera novaeangliae</i>	33	Pot ropes	Campbell (2011)
Sri Lanka	1985	Cetaceans	40 000	Gill-nets	Dawson (1991)
North-eastern USA, Atlantic Ocean	1977–93	<i>Delphinus delphis</i> , <i>Tursiops</i> spp.	230, 30	Trawl fisheries	Fertl and Leatherwood (1997)
North-west Africa	1980	<i>Delphinus delphis</i> and <i>Stenella</i> spp.	500–1000	Trawl fisheries	Fertl and Leatherwood (1997)
New Zealand	1978–95	<i>Delphinus delphis</i>	165	Trawl fisheries	Fertl and Leatherwood (1997)
Natal, South Africa	1994–2000	<i>Tursiops aduncus</i>	86	Shark control equipment	Natoli <i>et al.</i> (2008)
Pilbara, Western Australia	2002–12	<i>Tursiops aduncus</i>	500	Trawl fishery	Allen <i>et al.</i> (2014)

McPherson 2012). However, these trials were not successful in reducing the cetacean bycatch (Pirodda *et al.* 2016).

Shark control programs

A shark meshing program (Queensland Shark Control Program, QSCP) was introduced state wide along the popular beaches of Queensland, Australia, including the Gold Coast, in 1962. Similar shark control programs are in place in New South Wales (Australia) and South Africa. In New South Wales, bottom-set gill-nets are used on popular beaches from September to April and are removed from May to August each year, when there are fewer recreational swimmers (Green *et al.* 2009). In South Africa, shark nets have been in place in the nearshore waters of Durban since 1952 (KwaZulu–Natal Sharks Board 2013). On the Gold Coast, a combination of top-set nets and baited drum lines is used to minimise shark interactions with beachgoers. Bycatch species include finfish, stingrays, small sharks, turtles, dugongs, dolphins and whales (Gribble *et al.* 1998).

Environmental drivers

Entanglements may be driven by environmental conditions. *M. novaeangliae* are migratory animals that move annually from summer feeding grounds in the South and North poles to winter breeding grounds in the coastal waters of subtropical and tropical continents (Dawbin 1955; Jenner *et al.* 2001; Calambokidis *et al.* 2001). Studies by Friedlaender *et al.* (2006) and Dalla Rosa *et al.* (2012) correlated sightings of *M. novaeangliae* within temporal and spatial survey zones with environmental variables. *M. novaeangliae* in continental shelf waters were found to be colocated with the physical features that are also associated with zooplankton abundance, such as the distance from the ice edge, shelf slope and water depth (Friedlaender *et al.* 2006; Dalla Rosa *et al.* 2012). *M. novaeangliae* have also been found to use

currents such as the Antarctic Circumpolar Current (ACC); they appear to move with the southern boundary of the ACC as it meets the Upper Circumpolar Deep Water Current where an upwelling occurs (Tynan 1996). *M. novaeangliae* are thought to use the western boundary currents, such as the East Australian Current (EAC), in similar ways (Gendron and Urban 1993; Rosenbaum and Collins 2006). The presence of *M. novaeangliae* has also recently been linked with cooler waters and higher temperature gradients on the Gold Coast (Reinke *et al.* 2016).

In the southern, central and northern regions of New South Wales there are three broad variations in oceanographic conditions in the course of the EAC: the EAC intensification, the EAC zone of separation and the EAC's area of decline (Möller *et al.* 2011). There are three genetically different populations of *Delphinus delphis* that have been found to coincide with these variations in oceanographic conditions (Möller *et al.* 2011). A similar study on dolphin entanglement and environmental drivers in the central North Pacific by Ferrero *et al.* (2002) found three different dolphin species (*Lissodelphis borealis*, *Phocoenoides dalli* and *Lagenorhynchus obliquidens*) to be associated with distinct sea surface temperatures (SSTs). The combination of wind and swell and their effect on different areas of upwelling and productivity were also found to be significant (Ferrero *et al.* 2002).

Entanglements may also be driven by environmental conditions that increase biological productivity of the prey species of cetaceans. *M. novaeangliae* feed on krill (*Euphausia superba*) found in Antarctic waters during the summer (Department of Sustainability, Environment, Water, Population and Communities 2013). Migrating *M. novaeangliae* in Tasmania and Victoria (Australia) are thought to feed on a coastal species of Euphausiacea called *Nyctiphanes australis* (Department of Sustainability, Environment, Water, Population and Communities 2013). In temperate Queensland, they are thought to opportunistically

feed on *Sardinops sagax*, *Etrumeus teres*, *Scomber australasicus* and *Trachurus novaezelandiae* (Stockin and Burgess 2005). Different species of dolphins specialise on different prey species. For example, *D. delphis* and *Tursiops truncatus* are offshore species of dolphins that feed on squid and schooling fish (Rossbach and Herzing 1999; Department of the Environment and Energy 2017). *Tursiops aduncus* is an inshore ecotype of the bottlenose dolphin that feeds on bottom-dwelling animals and fish (Rossbach and Herzing 1999).

Aims

The aim of the present study was to define environmental conditions that occur when *M. novaeangliae* and *D. delphis* are entangled in the gill-nets of the QSCP on the Gold Coast.

Materials and methods

Study area

The Gold Coast is situated on the east coast of Australia at $\sim 28^{\circ}0'0.00''\text{S}$, $153^{\circ}24'0.00''\text{E}$. The coast curves in a mild embayment north of Point Danger, offering some shelter from the prevailing south-east swell. It is a highly urbanised region and a major international tourist destination.

The Gold Coast has had a varying amount of marine mammal entanglements in QSCP equipment since 1962. In total, 11 top-set gill-nets and 35 drum lines are used along the 37 km of coastline (Department of Agriculture and Fisheries 2012; Fig. 1). The gill-nets are set at depths of 10–12 m within 600–700 m from the shore and span 186 m parallel to the shore (Taylor *et al.* 2011; Department of Agriculture and Fisheries 2012).

Data

QSCP entanglements

The QSCP nets have been in the same locations since 1978. This ensures continuity for undertaking analysis on cetacean entanglements in the QSCP nets for the past 34 years. However, due to limitations in the QSCP contractor records and measurements from environmental monitoring instruments, the analysis herein is restricted to the period from 1990 to 2012 for *D. delphis* and from 1991 to 2012 for *M. novaeangliae*.

The QSCP cetacean entanglement records from 1968 to 2012 were provided by the Queensland DAF. Dolphin bycatch species were recorded as bottlenose, common, spinner, pantropical spotted, Indo-Pacific humpback and Irrawaddy. For data analysis, it was assumed that common dolphins were recorded as ‘dolphin’ by the QSCP contractors from 1989 to 1996 because all other dolphin species were identified by species name. From 1968 to 1983, whale entanglements were classified as ‘whale’ and there was a gap in QSCP whale entanglement records, with no records from 1984 to 1990. Therefore, analysis for the present study began from 1991 when whale entanglements were distinctly classified as ‘*M. novaeangliae*’.

The times of day for cetacean entanglements were recorded from 2005 to 2012. *M. novaeangliae* entanglements occurred between 0500 and 1500 hours, with the most common time of entanglement being 1000 hours. Therefore, unrecorded times of entanglements between 1991 and 2004 were defaulted to 1000 hours for analysis. Contractors are called out when a

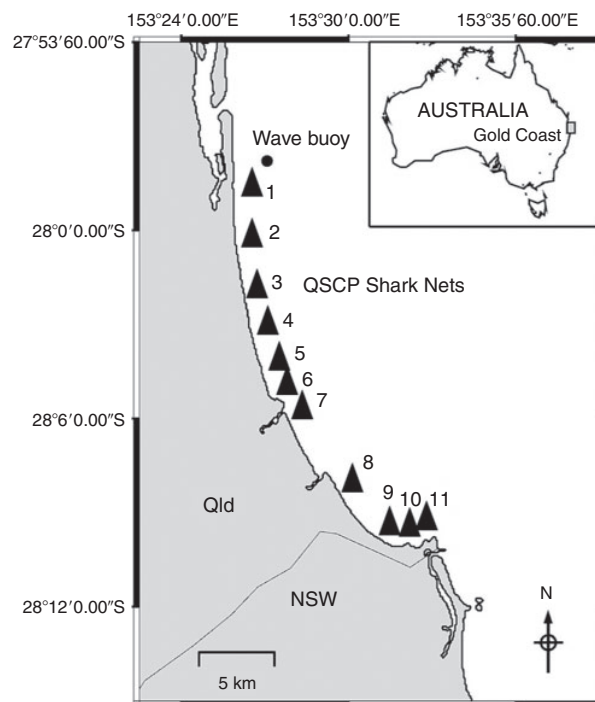


Fig. 1. Map showing the location of the Gold Coast (Qld, Australia). The black circle shows the position of the Gold Coast wave rider buoy and the black triangles show the positions of the Queensland Shark Control Program (QSCP) shark nets at Main Beach (1), Surfers Paradise (2), Kurrawa (3), Mermaid (4), Miami (5), Burleigh (6), Tallebudgera (7), Currumbin (8), Bilinga (9), Kirra (10) and Coolangatta (11). NSW, New South Wales.

M. novaeangliae entanglement occurs and data are collected per event; however, *D. delphis* entanglements in the nets mostly go unnoticed because dolphins become immobilised and drown underwater. The contractors check the nets every second day and dolphin entanglements are recorded during the service runs. The entanglement times recorded from 2005 to 2012 reflect the times that the contractors performed equipment checks. No time of day was recorded for dolphin entanglements from 1990 to 2004, therefore the trend of detected entanglements was continued with a default time that reflects the QSCP contractor route.

Finfish entanglements from 1990 to 2012 were supplied for correlation analysis with *M. novaeangliae* and *D. delphis* entanglements. Catches of the QSCP-targeted shark species that are over 2 m in length (*Sphyrna mokarran*, *Carcharhinus leucas*, *Galeocerdo cuvier* and *Carcharodon carcharias*) were also supplied for correlation analyses between 2001 and 2013 to investigate whether similar environmental conditions existed between animal entanglements.

Environmental data

Wind speed and direction

Wind data were provided by the Bureau of Meteorology from the automatic weather station at the Gold Coast Seaway. Environmental data were categorised into groups that represent an environmental condition instead of single values, which

reduced the data points provided by monitoring instruments. For example, the Beaufort wind scale was used to categorise wind speed. A cetacean entanglement can now be reported as having occurred in 'light winds (between 12 and 19 km h⁻¹)' instead of at 15.4 km h⁻¹. To analyse the effects of upwelling and downwelling events on entanglements, wind direction was vector averaged from 1991 to 2012 for *M. novaeangliae* entanglements and from 2001 to 2008 for *D. delphis* entanglements. The wind-driven horizontal movement of water can induce vertical movements in surface water, known as downwelling, that affects water clarity (Garrison 2007). The opposite, upwelling events, can also affect water clarity and increase the biological productivity of the area, which may attract cetacean species (Garrison 2007).

Wave power

The power associated with the wave climate (kW m⁻¹) was calculated by squaring the significant wave height (Hs), multiplying by the peak wave period (Tp) and then multiplying by a factor of 0.49 (Behrens *et al.* 2012):

$$\text{Power} = H_s^2 \times T_p \times 0.49$$

Wave power was binned into 5 kW m⁻¹ categories to represent the range of wave climates during cetacean entanglement events on the Gold Coast. The wave energy affecting the gill-nets is assumed to move the nets in an oscillatory motion and the larger the wave power, the greater the movement.

Rainfall

Hourly rainfall data measured at the Gold Coast Seaway were provided by the Bureau of Meteorology. The rate of precipitation was categorised using the Bureau of Meteorology's standard categories for routine weather forecasts (<http://www.bom.gov.au/info/wwords/>, accessed 29 March 2017), namely dry (0 mm h⁻¹), drizzle (0.01–0.9 mm h⁻¹), light to moderate rain (1–6 mm h⁻¹) and heavy to violent rain (>6.2 mm h⁻¹). Precipitation rates can affect run-off from the catchment and therefore affect water clarity and quality (Garrison 2007).

Moon phase and tides

Moon phase data were derived from the software program QuickPhase Pro (MoonConnection.com, Park City, UT, USA, see <http://www.moonconnection.com/quickphase/>, accessed 29 March 2017). The percentage full value for each Moon phase was categorised as either spring (75–100 and 0–25%) or neap (25–75%). Tidal heights have been correlated with fish migrations from estuaries (Boehlert and Mundy 1988) and the lunar light intensity has been correlated with the vertical migration of zooplankton and fish species (Benoit-Bird *et al.* 2009; Elwen *et al.* 2009).

East Australian Current

The EAC is a narrow, fast and deep western boundary current that transports warm waters from the equator polewards (Garrison 2007). It is the most energetic feature in the subtropical Pacific region (Schiller *et al.* 2008). The offshore distance to the centre of

the EAC jet was provided by CSIRO from the BLUElink Ocean Reanalysis model from 1992 to 2008 (Schiller *et al.* 2008). The location of the EAC jet, defined as the region of maximum southerly current, changes frequently due to meandering and the passage of meso-scale eddies and the location is thought to affect the coastal waters of the Gold Coast (D. Kobashi, L. Hughes, R. Robertson and R. Tomlinson, unpubl. data).

SST and wave height

Wave and SST data were provided by the Queensland Department of Science, Information Technology and Innovation (DSITI) from the Gold Coast wave-rider buoy, which is moored at a depth of 17 m at Main Beach (27°57'54.84"S, 153°26'30"E). This wave-rider buoy provides wave parameters at a similar depth and location to the shark nets. Environmental data were categorised into groups that represent sea conditions instead of single values in order to reduce the data points provided by monitoring instruments. For example, the Bureau of Meteorology's sea height scale was used to categorise wave heights (see <http://www.bom.gov.au/info/wwords/>). A cetacean entanglement can now be reported as having occurred in 'smooth seas (between 0.1 and 0.5 m)' instead of in a 0.3-m swell.

SST has been correlated with the habitat ranges and survival tolerances of many marine animal species (Krebs 2009). The wave-rider buoy recorded SST data with half-hourly measurements from 2008 to 2012. From 2001 to 2007, data were sourced from the BLUElink Ocean Reanalysis model (CSIRO) providing daily SST data.

Statistical analysis

Due to the zero-inflated and binary distribution of the dependent variable (entanglement and no entanglement) in the present study, a Chi-Square analysis was selected. Analysis of categorical data was viewed as the most appropriate way to investigate environmental conditions that are hypothesised to correlate with cetacean entanglements.

The counts of both *M. novaeangliae* and *D. delphis* entanglements were tallied for the categories of wind speed and direction, wave height and power, rainfall and moon phase. The α -level for the Chi-Square tests was set to 0.05. The prevailing conditions experienced on the Gold Coast were weighted against each environmental variable to take into the analysis the probability of each category occurring. This was done by categorising and tallying the values from the environmental monitoring instruments from May to November (from 1991 to 2012) to obtain data climates for *M. novaeangliae* and from January to December (from 1990 to 2012) to obtain data climates for *D. delphis*. The probability of each category occurring was then multiplied by the number of cetacean entanglements in each category to obtain a weighted value. Correlation analysis was undertaken for the variables of the position of the EAC, SST, finfish and targeted shark catches with cetacean entanglement dates.

Wind speed, wind direction, wave height, wave power and rainfall were lagged (24, 48 and 72 h prior to entanglement) because these environmental drivers may affect oceanic properties over multiple days and therefore possibly cetacean entanglements. Wind speed, wave height and wave power were

averaged for the 24- and 48-h time periods before entanglement. Wind direction was vector averaged by first converting each record from magnitude in kilometres per hour and direction in degrees clockwise from 0°N to its corresponding north–south and east–west vector components. The north–south component of wind direction was used in this report to determine the effects of wind-induced coastal upwelling (north component) and downwelling (south component; Garrison 2007). Time lags were used for rainfall to include high rainfall events occurring 72 h before an entanglement. The largest rate of precipitation that occurred in the different time lags was used to select the category.

Results

Occurrence of entanglements

The results reported herein are presented as percentages of total entanglements due to restrictions and sensitivity of the data. The catch data for *D. delphis* met the assumptions for a Chi-Square probability analysis in all tests. However, the *M. novaeangliae* catch data did not meet the expected cell frequency assumption for a Chi-Square probability test for

net location and years that entanglements occurred (De Veaux *et al.* 2008).

The highest number of *M. novaeangliae* entanglements occurred in September each year and the lowest number of entanglements occurred in May and November, with a significant difference between the months ($\chi^2_5 = 14.74$, $P < 0.025$) (Table 2). A comparison between the northern (April–July) and southern (August–November) migration seasons for *M. novaeangliae* revealed a significant difference between the number of entanglements, with a 95% greater chance that entanglements occurred during the southern migration route ($\chi^2_1 = 4.3$, $P < 0.05$).

Entanglements of dolphins (*D. delphis*) varied significantly between years ($\chi^2_{22} = 39.12$, $P < 0.025$), with the highest entanglements in 2010 and the lowest entanglements occurring in 1997 and 1999. There was also a significant difference between the months that entanglements occurred ($\chi^2_{11} = 20.12$, $P < 0.05$), with the highest entanglements occurring in June and the lowest occurring in the summer months of December, January and February. The highest number of dolphin entanglements occurred in the Main Beach net, whereas the lowest number of entanglements occurred in the Kurrawa and Mermaid Beach nets (Fig. 1) ($\chi^2_{10} = 18.39$, $P < 0.05$).

Table 2. Summary of the effects of environmental drivers on *Megaptera novaeangliae* and *Delphinus delphis* entanglements in the nets of the Queensland Shark Control Program
NA, not available; Hs, significant wave height

Variable	Time lags	Species	Chi-Square value	P-value	Category with greatest number of entanglements	Category with lowest number of entanglements
Month	NA	<i>Megaptera novaeangliae</i>	$\chi^2_5 = 14.74$	<0.025	September	May and November
	NA	<i>Delphinus delphis</i>	$\chi^2_{11} = 20.12$	<0.05	June	December, January and February
Net location	NA	<i>Delphinus delphis</i>	$\chi^2_{10} = 18.39$	<0.05	Main Beach	Kurrawa and Mermaid
Year	NA	<i>Delphinus delphis</i>	$\chi^2_{22} = 39.12$	<0.025	2010	1997 and 1999
Wave height (Hs)	Incidental	<i>Megaptera novaeangliae</i>	$\chi^2_2 = 23.8$	<0.005	0.5–1.25 m	0.1–0.5 m
	24 h	<i>Megaptera novaeangliae</i>	$\chi^2_2 = 50.15$	<0.005		
	48 h	<i>Megaptera novaeangliae</i>	$\chi^2_2 = 14.55$	<0.005		
	Incidental	<i>Delphinus delphis</i>	$\chi^2_2 = 35.7$	<0.005		
Wave power	24 h	<i>Delphinus delphis</i>	$\chi^2_2 = 55.4$	<0.005	0.5–1.25 m	0.1–0.5 m
	Incidental, 24 h	<i>Megaptera novaeangliae</i>	NA		0–5 kW m ⁻¹	15–20 kW m ⁻¹
	48 h	<i>Megaptera novaeangliae</i>	NA		0–5 kW m ⁻¹	10–15 kW m ⁻¹
	Incidental, 24 h	<i>Delphinus delphis</i>	NA		0–5 kW m ⁻¹	20–25 kW m ⁻¹
Wind speed	48 h	<i>Delphinus delphis</i>	NA		0–5 kW m ⁻¹	25–35 kW m ⁻¹
	6 h	<i>Megaptera novaeangliae</i>	$\chi^2_3 = 15.9$	<0.005	12–19 km h ⁻¹	30–39 km h ⁻¹
	12 h	<i>Megaptera novaeangliae</i>	$\chi^2_3 = 40.3$	<0.005		
	24 h	<i>Megaptera novaeangliae</i>	$\chi^2_4 = 97.5$	<0.005		
Rainfall	Incidental	<i>Delphinus delphis</i>	$\chi^2_4 = 23.8$	<0.005	6 and 11 km h ⁻¹	1–5 km h ⁻¹
	24 h	<i>Delphinus delphis</i>	$\chi^2_3 = 69.7$	<0.005	12–19 km h ⁻¹	30–39 km h ⁻¹
	48 h	<i>Delphinus delphis</i>	$\chi^2_5 = 124.3$	<0.005	12–19 km h ⁻¹	1–5 and 40–50 km h ⁻¹
	24 h	<i>Megaptera novaeangliae</i>	$\chi^2_3 = 23.4$	<0.005	1–6 mm	>6.2 mm
	48 h	<i>Megaptera novaeangliae</i>	$\chi^2_3 = 20.7$	<0.005	0 mm	>6.2 mm
	72 h	<i>Megaptera novaeangliae</i>	$\chi^2_3 = 19.9$	<0.005	0–1 mm	>6.2 mm
Moon phase	NA	<i>Delphinus delphis</i>	$\chi^2_1 = 17.63$	<0.005	Spring tides	Neap tides

Environmental drivers

Wave height

Cetacean entanglements occurred in three of the four sea state categories, namely smooth (0.1–0.5 m), slight (0.5–1.25 m) and moderate (1.25–2.5 m). Entanglements in the rough (2.5–4 m) category did not occur. The percentage of *M. novaeangliae* entanglements were weighted against the wave climate on the Gold Coast during the whale migration between May and November each year (smooth, 4%; slight, 72%; moderate, 22%). The differences in *M. novaeangliae* entanglements in relation to wave height categories at the three different time periods were all found to be statistically significant (Fig. 2). The highest number of entanglements occurred in the 0.5–1.25-m category and the lowest number of entanglements occurred in the 0.1–0.5-m category.

Because *D. delphis* entanglements occurred throughout the entire year, the percentage of entanglements was weighted against the wave climate on the Gold Coast across all months (smooth, 3%; slight, 66%; moderate, 29%). The differences in entanglements in relation to wave height categories were significant for two of the time periods analysed, namely the incidental ($\chi^2_2 = 35.7, P < 0.005$) and the 24 h before entanglements ($\chi^2_2 = 55.4, P < 0.005$), with the slight category having the greatest entanglements (Fig. 2). There was no significant difference found for the 48 h before the entanglement time period ($\chi^2_1 = 1.04, P > 0.10$).

Wave power

M. novaeangliae entanglements mostly occurred when the wave power was between 0 and 5 kW m⁻¹ and did not occur once the wave power was greater than 20 kW m⁻¹, coinciding with the dominant wave power on the Gold Coast from May to November (0–5 kW m⁻¹ for 59% of the time between 1991 and 2012). Results were similar for *D. delphis* entanglements, for which most entanglements occurred when the wave power was below 10 kW m⁻¹. The average annual wave power on the Gold Coast from January to December was 0–5 kW m⁻¹ (53% of the time) and 5–10 kW m⁻¹ (28% of the time).

Wind speed

The most *M. novaeangliae* entanglements occurred when the wind speed was between 12 and 19 km h⁻¹, whereas the lowest number of entanglements occurred for wind speeds <5 or >30 km h⁻¹ (Fig. 3). From 1991 to 2012, Gold Coast wind speeds were in the 12–19 km h⁻¹ category 33% of the time. The periods 6 h before entanglement ($\chi^2_3 = 15.9, P < 0.005$), 12 h before entanglement ($\chi^2_3 = 40.3, P < 0.005$) and 24 h before entanglement time ($\chi^2_4 = 97.5, P < 0.005$) were all statistically significant, which means there was a difference in the number of entanglements across the categories. The degrees of freedom were higher for the 48 h before an entanglement time period because the wind speed reached 40–50 km h⁻¹, whereas for the two other time periods the highest wind speed reached was 30–39 km h⁻¹.

The highest number of *D. delphis* entanglements occurred between wind speeds of 6 and 11 km h⁻¹ ($\chi^2_4 = 23.8, P < 0.005$), coinciding with the average wind speed from January to December (<19 km h⁻¹ for 65% of the time). However, the

time periods 24 h before entanglement ($\chi^2_3 = 69.7, P < 0.005$) and 48 h before entanglement ($\chi^2_5 = 124.3, P < 0.005$) had the highest entanglement rates occurring in the 12–19 km h⁻¹ category (Fig. 4).

Wind-induced vertical circulation and Moon phase

There was no significant difference detected between the wind-induced vertical circulation categories of upwelling and downwelling on the Gold Coast in relation to the incidence of

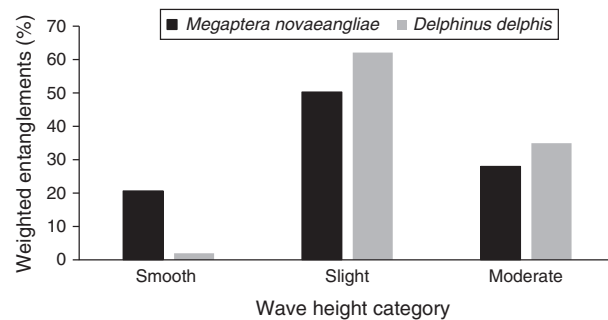


Fig. 2. Percentage of weighted *Megaptera novaeangliae* and *Delphinus delphis* entanglements in the smooth (0.1–0.5 m), slight (0.5–1.25 m) and moderate (1.25–2.5 m) wave height categories using the average wave height 24 h before the entanglement occurred.

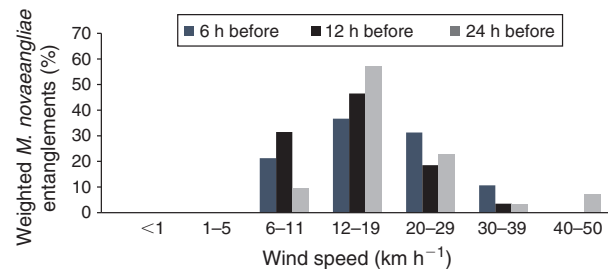


Fig. 3. Percentage of weighted *Megaptera novaeangliae* entanglements in each wind speed category from 1991 to 2012 using the average wind speed 6, 12 and 24 h before the entanglement occurred.

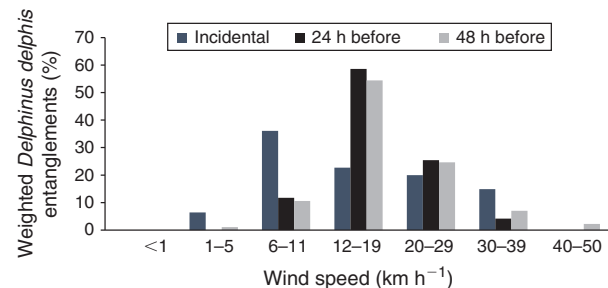


Fig. 4. Percentage of weighted *Delphinus delphis* entanglements in each wind speed category from 1990 to 2012 using the wind speed at the time of entanglement and averaged wind speed 24 and 48 h before the entanglement occurred.

M. novaeangliae or *D. delphis* entanglements. All three time periods analysed were not significant for either species.

There was no significant difference between the occurrence of spring and neap tides and *M. novaeangliae* entanglements ($\chi^2_1 = 2.63$, $P > 0.10$) on the Gold Coast. However, there was a significant difference between Moon phase for *D. delphis* entanglements, with the probability of entanglements being 99.5% greater with spring tides ($\chi^2_1 = 17.63$, $P < 0.005$) than neap tides.

Rainfall

Entanglements of *M. novaeangliae* in the QSCP are rare occurrences when the precipitation rate is greater than 6.2 mm h^{-1} , which represents heavy to violent rainfall events. The periods 24 h before an entanglement event ($\chi^2_3 = 23.4$, $P < 0.005$), 48 h before an entanglement event ($\chi^2_3 = 20.7$, $P < 0.005$) and 72 h before an entanglement event ($\chi^2_3 = 19.9$, $P < 0.005$) all showed a significant difference between each of the categories after weighting against the rainfall climate on the Gold Coast. There were no statistically significant differences in *D. delphis* entanglements between rainfall rates.

Position of the EAC

M. novaeangliae entanglements have not occurred while the maximum velocity of the EAC is located further east than $154^\circ 30' 0.00''\text{E}$ (Fig. 5). The average position of the EAC from July to October was $154^\circ 0' 0.00''\text{E}$. *M. novaeangliae* entanglements occurred 72.7% of the time when the EAC was west (shoreward) of its average position.

D. delphis entanglements between 1992 and 2008 occurred 90.9% of the time when the EAC was west of $154^\circ 30' 0.00''\text{E}$

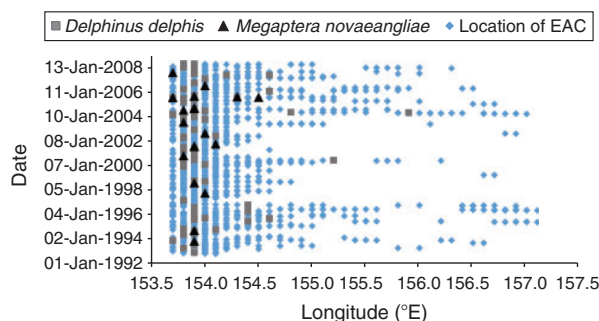


Fig. 5. Daily longitude location of the East Australian Current (EAC) between 1992 and 2008 and the entanglement dates of *Delphinus delphis* and *Megaptera novaeangliae*.

(Fig. 5). The average position of the EAC from January to December is the same as from July to October at 154°E . Most *D. delphis* entanglements (79.2%) occurred when the EAC was west of the average position.

Sea surface temperature

Comparison of *M. novaeangliae* and *D. delphis* entanglement with SST revealed no significant relationships. However, entanglements of *M. novaeangliae* and *D. delphis* did not occur below an SST of 18.6°C between 2001 and 2007. The lowest SST recorded between 2001 and 2007 (BLUElink Ocean Reanalysis model; CSIRO) was 17.6°C in September of 2007. In 2001, the SST did not fall below 19.6°C and, in that year, a high *M. novaeangliae* entanglement rate was recorded (10.5% of the total entanglements between 1990 and 2012).

Finfish and elasmobranch catches in the QSCP

No significant matches between finfish catches and cetacean entanglements were found. Sharks that were caught on the same day were classified as a 'direct match' and sharks that were caught within 3 days before or after a cetacean entanglement were classified as a 'close match'. Most *M. novaeangliae* and *D. delphis* entanglements (81 and 87% respectively) did not coincide with dates when there were sharks caught that were over 2 m in length (Table 3).

Discussion

Sea state

Both *M. novaeangliae* and *D. delphis* entanglements were proven statistically to occur during calm sea states (Table 2). The QSCP nets are in nearshore waters 600–700 m offshore and in depths of 10–12 m. Calm conditions may be preferred by *M. novaeangliae* and *D. delphis* for resting in nearshore waters, thereby increasing their chances of entanglement (Dudzinski 1996; Hawkins and Gardside 2008; Franklin 2012; Meynecke *et al.* 2013).

The QSCP gill-nets may also be stationary in calm conditions, which could affect the sight and sound perceptions of cetaceans to detect the nets (Erbe and McPherson 2012). No sound would be emitted from the gill-nets when the nets are stationary and when there is limited wave breaking near the nets in calm conditions. As wave height increases, there would be an increasing vertical movement of the gill-nets (Garrison, 2007). The energy affecting the nets would be assumed to move the nets in an oscillatory motion, with the larger the wave power the greater the movement. The visual and auditory capacity of *M. novaeangliae* and *D. delphis* to perceive objects at distances is unknown and further research is needed in this area. The catch

Table 3. Percentage of entanglements between marine mammals and dangerous species of sharks over 2 m in length in Queensland Shark Control Program nets

	Direct entanglement date match	Close entanglement date match	No entanglement date match
<i>Megaptera novaeangliae</i>	4%	15%	81%
<i>Stenella longirostris</i> , <i>S. attenuata</i> , <i>Sousa chinensis</i> and <i>Tursiops</i> spp.	6%	8%	86%

data provided for the present study confirmed that the alarms have had no effect on reducing entanglements in the QSCP nets. A study from New South Wales, Australia, similarly found no effect of whale alarms on *M. novaeangliae* when observed during their northern migration (Harcourt *et al.* 2014).

Migrations

Seasonal variations in the number of entanglements were evident for both marine mammal species. A greater percentage of entanglements of *M. novaeangliae* occurred during their southern migration (August–October) compared with their northern migration (June–July). During the southern migration, *M. novaeangliae* pod compositions have been observed to change from singletons to mother and calf pairs in Hervey Bay (Franklin 2012) and the Gold Coast (Meynecke *et al.* 2013). On the southern migration route, *M. novaeangliae* have poorer body condition and they may seek out sheltered coastal zones to conserve their energy reserves (Wiley *et al.* 1995; Franklin 2012; Meynecke *et al.* 2013). A longer residence time in Gold Coast waters during the southern migration leads to a greater potential of becoming entangled in nets.

The body lengths of entangled *M. novaeangliae* that have been recorded were between 4 and 12 m, with the average body length of 7.5 m. This suggests that predominately juvenile *M. novaeangliae* have been entangled in the Gold Coast waters, because adult *M. novaeangliae* measure between 13 and 18 m in length (Franklin 2012). The behaviour of juvenile *M. novaeangliae* pods differs from that of adults, with juveniles forming active pods full of energetic, adolescent, non-agonistic social behaviour (Wiley *et al.* 1995; Franklin and Woolley 2000). The behaviour of juvenile *M. novaeangliae* may be a factor in their entanglement in the QSCP nets.

D. delphis were the most abundant dolphin species caught in the nets. Entanglements did occur for *Stenella longirostris*, *Stenella attenuata*, *Sousa chinensis* and *Tursiops* spp. on the Gold Coast, but these events were rare. The highest percentage of *D. delphis* entanglements occurred in June. This is the start of the spawning season for the potential prey species of *D. delphis*, namely *Pomatopus saltatrix*, *S. sagax* and *E. teres*, which migrate into the cool coastal waters of the Gold Coast (Ward *et al.* 2003). Predation into nearshore waters may increase the likelihood of entanglements of transient *D. delphis*.

Net locations near estuaries had the highest number of *D. delphis* entanglements (Gold Coast Seaway, Tallebudgera Creek, Currumbin Creek and Tweed River). The Main Beach net had the greatest percentage of *D. delphis* entanglements, and this net is the closest of all the QSCP nets to the large estuary network of the Broadwater. Food abundance has been proven to be higher near estuaries because they provide the habitat needed for the life cycle of many prey species (Connell and Gillanders 2007). A survey of the populations of dolphins on the Gold Coast has shown that they are mostly involved in feeding behaviour 60% of the time and are mostly present in estuarine and nearshore waters (E. Hawkins, L. Brooks and A. Peterson, unpubl. data, 2011).

There was a significant relationship between spring tides and *D. delphis* entanglements, and this also may be related to prey movements and feeding behaviour of *D. delphis*. The lunar light intensity associated with spring tides affects the vertical migration

of prey (Benoit-Bird *et al.* 2009; Elwen *et al.* 2009) and the increased tidal flux of an estuary system that occurs during spring tides is used by prey species for dispersal, whereas other species use it for predation (Boehlert and Mundy 1988). In surveys of *S. longirostris* in Hawaii, *L. obscurus* in New Zealand and *Cephalorhynchus heavisidii* in South Africa, Benoit-Bird *et al.* (2009) found that the relative abundance of dolphins increased with lunar illumination.

Water quality

M. novaeangliae entanglements mostly occurred when the rainfall rate was $\leq 6 \text{ mm h}^{-1}$, and entanglements rarely occurred during periods of heavy to violent rainfall. Reduced salinity, increased turbidity and increased pollutants from run-off after heavy to violent rainfall events may cause *M. novaeangliae* to move further offshore (OZCoasts 2012). Heavy to violent rain events are also associated with stronger wind speeds and rough to phenomenal sea states, instead of calm conditions, when most entanglements have occurred (Tarbuck *et al.* 2009).

A similar study that analysed environmental drivers of *C. leucas* catches in the QSCP also found statistically significant correlations with rainfall (Werry 2010). In that study, there was an increase in adult *C. leucas* caught on baited drumlines after an aggregate of 300-mm rainfall had occurred in the 8 days before the individuals were caught (Werry 2010). This suggests active feeding, where larger bull sharks may be moving from depths $> 25 \text{ m}$ to feed in freshwater plumes extruding from river mouths after rainfall events (Werry 2010).

East Australian Current

M. novaeangliae entanglements did not occur when the EAC was further east than $154^{\circ}30'0.00''\text{E}$. This observation suggests that the EAC may affect the distribution and abundance of *M. novaeangliae* in Gold Coast waters during their migration seasons. *M. novaeangliae* may be using coastal fronts and prefer areas of high temperature gradients for navigation. Preferences for warmer waters were shown for the Antarctic Circumpolar Current by Tynan (1996), whereby *M. novaeangliae* were found to be using a warmer subsurface section of the current with high krill abundance.

The Gold Coast region is considered a low-production ecosystem ($< 150 \text{ g C m}^{-2} \text{ year}^{-1}$; Aquarone *et al.* 2012). The highest productivity occurs in localised coastal blooms that occur as a result of wind- and current-driven upwelling events; however, these events are not large enough to support a large demersal fishery (Aquarone *et al.* 2012). The vast bulk of commercial fishery landings are near the narrow continental shelf (Aquarone *et al.* 2012). There may be a preference for *D. delphis* to come closer to shore when the EAC is intruding into coastal waters. This phenomenon was observed for six species of cetaceans in the Gibraltar Strait in the Mediterranean Sea, where there is a surface inflow of Atlantic waters and a deep outflow of Mediterranean waters (de Stephanis *et al.* 2008).

Conclusion

The present study addressed the relationship between environmental drivers and cetacean entanglements in the QSCP nets. Calm conditions were determined to be an important factor that

influenced entanglements of both species. Seasonal differences were apparent that could be related to the migration of the marine mammals or the migrations of their prey species. Water quality may also be a factor affecting the abundance of *M. novaeangliae* in coastal waters. The movements of the EAC may also be an underestimated factor affecting marine mammals on the east coast of Australia.

Although the assumptions for a Chi-Square probability statistical test were met in the present study, the small number of *M. novaeangliae* entanglements recorded limited the robustness of the correlations obtained. In addition, the high variability of environmental factors limits the outcomes of the present study. Further statistical analyses using a generalised linear model with a Poisson distribution and a log canonical link function could be used to show the possibilities of other scenarios and interactions between the variables (Dobson 2002). This would use the data in a continuous form instead of the method chosen in the present study, namely using data as discrete categorical variables.

To mitigate entanglements, management interventions need to consider environmental conditions for *M. novaeangliae* from July to October when there are calm conditions, $<6 \text{ mm h}^{-1}$ rainfall and when the EAC is closer to shore than $154^{\circ}30'0.00''\text{E}$. To mitigate the effects of nets on *D. delphis* entanglements between March and November, managers should pay particular attention to the month of June, when there are calm conditions, spring tides and the EAC is closer to shore than the average position of 154°E . Management actions could include having marine mammal spotters, deploying acoustic listening stations or removing the nets during high-risk conditions.

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